

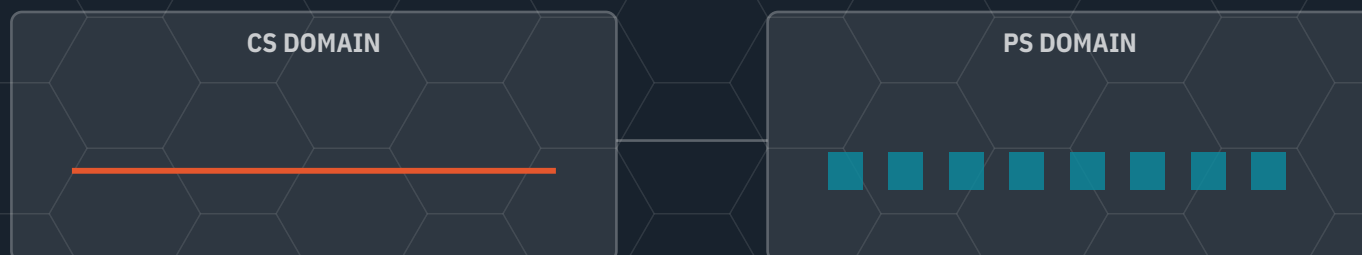
# Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

## Module 3

## 2G/3G Core Networks

شبکه‌های هسته 2G/3G



### Lecture Notes

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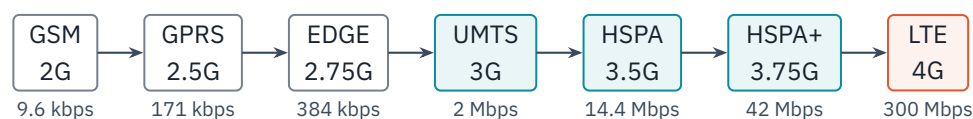
## 3.1 Evolution from 2G to 3G — The Driving Forces

### Why Did We Need 3G?

The transition from 2G (GSM) to 3G (UMTS) was driven by fundamental market and technology demands:

1. **Exploding Data Demand:** SMS and basic WAP browsing were insufficient. Users wanted email, web browsing, and file downloads on mobile devices.
2. **Multimedia Services:** Video calling, streaming audio/video, and rich content required bandwidth far beyond GSM's 9.6 kbps (or even GPRS's theoretical 171 kbps).
3. **Higher Data Rates:** The ITU's IMT-2000 vision targeted:
  - 144 kbps for high mobility (vehicular)
  - 384 kbps for pedestrian
  - 2 Mbps for stationary/indoor
4. **Global Roaming:** A unified worldwide standard (unlike fragmented 2G: GSM vs CDMA vs PDC).
5. **Spectrum Efficiency:** WCDMA's spreading codes allow more users per cell with graceful degradation.

### The Evolution Path



**Figure 3.1** Generations and their peak data rates

## 3.2 UMTS Architecture Overview

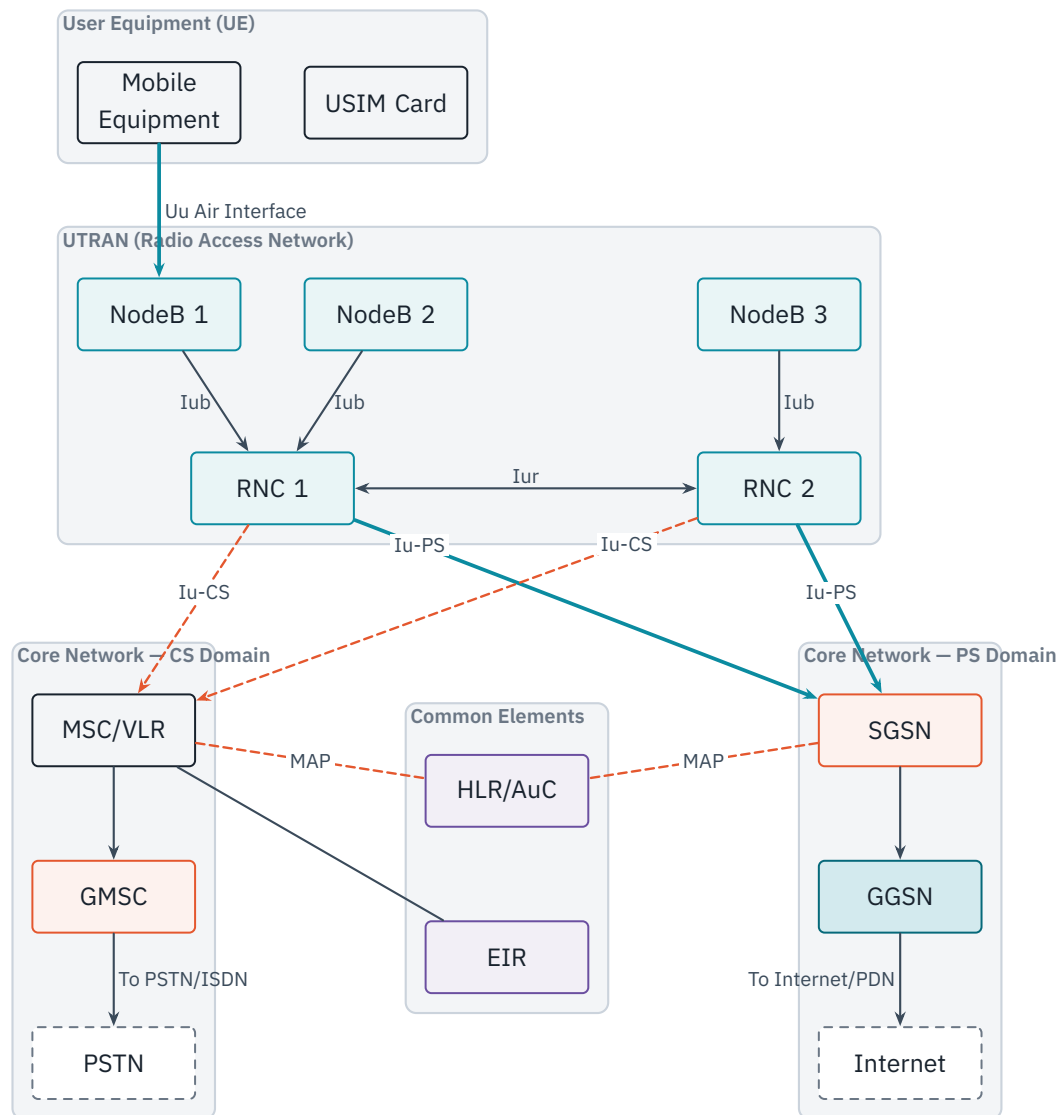
UMTS (Universal Mobile Telecommunications System) is built on two major domains:

- **UTRAN** (UMTS Terrestrial Radio Access Network) — the radio/air interface side
- **Core Network (CN)** — the intelligence and connectivity backbone

### ■ Key Design Principles

- Backward compatibility with GSM/GPRS core
- Separation of radio access from core network (allowing independent evolution)
- Split core into Circuit-Switched (CS) and Packet-Switched (PS) domains
- Support for simultaneous voice and data

## ■ UMTS Architecture Diagram



**Figure 3.2** UMTS architecture: UTRAN with the CS and PS core domains

## 3.3 UTRAN: NodeB and RNC

### NodeB (Base Station)

The NodeB is the 3G equivalent of the GSM BTS (Base Transceiver Station):

| Aspect                | Function                                               |
|-----------------------|--------------------------------------------------------|
| Primary Role          | Radio transmission/reception over the Uu air interface |
| Spreading/Despreading | Applies WCDMA spreading codes                          |
| Power Control         | Inner-loop power control (fast, 1500 Hz)               |
| Soft Handover         | Supports macro-diversity combining                     |
| Channel Coding        | Encoding/decoding of transport channels                |
| Interface             | Connected to RNC via Iub interface (ATM or IP-based)   |

**Key Difference from GSM BTS:** NodeB has MORE intelligence — it handles fast power control and soft handover combining locally, reducing latency.

### RNC (Radio Network Controller)

The RNC is the 3G equivalent of the GSM BSC, but with significantly more responsibilities:

| Aspect                    | Function                                         |
|---------------------------|--------------------------------------------------|
| Radio Resource Management | Admission control, code allocation, load control |
| Handover Decision         | Soft/softer/hard handover management             |
| Outer-Loop Power Control  | SIR target adjustment                            |
| Ciphering/Integrity       | Security functions for user data and signaling   |
| Protocol Termination      | RRC (Radio Resource Control) protocol endpoint   |
| Macro-Diversity Combining | Combines signals from multiple NodeBs            |
| Segmentation/Reassembly   | RLC layer processing                             |

### RNC Roles (One RNC Can Play Multiple Roles)

- **Serving RNC (S-RNC):** Maintains the RRC connection with UE, performs outer-loop power control
- **Drift RNC (D-RNC):** Provides radio resources when UE moves but S-RNC doesn't change
- **Controlling RNC (C-RNC):** Controls the logical resources of its NodeBs

## UTRAN Interfaces

| Interface | Connects    | Protocol Stack      | Purpose                             |
|-----------|-------------|---------------------|-------------------------------------|
| Uu        | UE ↔ NodeB  | WCDMA air interface | Radio transmission                  |
| Iub       | NodeB ↔ RNC | NBAP over ATM/IP    | NodeB control and user data         |
| Iur       | RNC ↔ RNC   | RNSAP over ATM/IP   | Inter-RNC soft handover, relocation |
| Iu-CS     | RNC ↔ MSC   | RANAP over ATM      | Circuit-switched traffic            |
| Iu-PS     | RNC ↔ SGSN  | RANAP over IP       | Packet-switched traffic             |

### 3.4 3G Core Network: CS Domain + PS Domain

The 3G core network inherits the GSM/GPRS split architecture, maintaining two parallel domains:

#### Circuit-Switched (CS) Domain

Handles real-time voice calls and video telephony (same as GSM core, upgraded for 3G).

| Element                           | Role                                                                |
|-----------------------------------|---------------------------------------------------------------------|
| MSC (Mobile Switching Centre)     | Call routing, handover between RNCs, call control signaling         |
| VLR (Visitor Location Register)   | Temporary storage of subscriber data for visitors in the MSC area   |
| GMSC (Gateway MSC)                | Interface to external networks (PSTN, ISDN), handles incoming calls |
| HLR (Home Location Register)      | Permanent subscriber database (shared with PS domain)               |
| AuC (Authentication Centre)       | Stores authentication keys (Ki), generates security vectors         |
| EIR (Equipment Identity Register) | IMEI blacklist/whitelist for stolen device blocking                 |

CS Domain Call Flow:



Figure 3.3 CS domain call flow

## ■ Packet-Switched (PS) Domain

Handles all data services (web, email, streaming) — evolved from GPRS.

| Element                          | Role                                                                                    |
|----------------------------------|-----------------------------------------------------------------------------------------|
| SGSN (Serving GPRS Support Node) | Packet routing, mobility management, session management, ciphering                      |
| GGSN (Gateway GPRS Support Node) | Interface to external packet networks (Internet), IP address allocation, billing anchor |

PS Domain Data Flow:

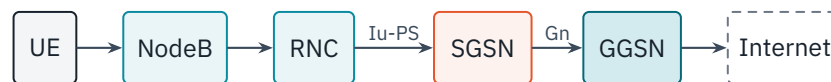


Figure 3.4 PS domain data flow

## ■ Why Two Domains?

The split CS/PS approach was a **pragmatic engineering decision**:

1. **Legacy Compatibility:** Voice infrastructure (SS7, PSTN interconnect) was mature and reliable — no reason to rebuild it
2. **QoS Guarantees:** Circuit-switching provides deterministic delay for voice (guaranteed 20ms frame delivery)
3. **Incremental Deployment:** Operators could add PS domain without disrupting existing voice revenue
4. **Risk Mitigation:** IP networks in 2000 couldn't guarantee voice quality — packet loss and jitter were real problems

## 3.5 The Iu Interface

The Iu interface is the critical boundary between UTRAN and the Core Network.

### ■ Iu-CS (Circuit-Switched)

- **Transport:** ATM (AAL2 for user plane, AAL5 for control plane)
- **Control Plane:** RANAP → SCCP → M3UA → SCTP (or MTP3b)
- **User Plane:** Iu UP protocol (framing for AMR voice codec frames)
- **Purpose:** Carries voice calls and video telephony

## Iu-PS (Packet-Switched)

- **Transport:** IP-based (GTP-U for user plane)
- **Control Plane:** RANAP → SCCP → M3UA → SCTP
- **User Plane:** GTP-U (GPRS Tunneling Protocol - User plane)
- **Purpose:** Carries all packet data (web, email, streaming)

## RANAP (Radio Access Network Application Part)

Key procedures over Iu:

- **RAB (Radio Access Bearer) Assignment:** Sets up the end-to-end bearer
- **Relocation:** Moves Serving RNC role to another RNC (hard handover)
- **Paging:** Core network pages UE through UTRAN
- **Security Mode:** Activates ciphering and integrity protection
- **Direct Transfer:** Passes NAS messages transparently (e.g., MM, CC, SM)

## 3.6 Release 99 vs Release 5+ (HSPA Evolution)

### Release 99 (R99) – Original UMTS

- **Downlink:** Up to 384 kbps (dedicated channels)
- **Uplink:** Up to 384 kbps
- **Channel Type:** Dedicated Channel (DCH) — one code per user
- **Scheduling:** Done at RNC level (slow, ~100ms TTI)
- **Power Control:** Primary mechanism for link adaptation
- **Limitation:** Inefficient for bursty data (keeps resources allocated even during silence)

## Release 5 – HSDPA (High Speed Downlink Packet Access)

| Feature          | Improvement                                         |
|------------------|-----------------------------------------------------|
| Peak Rate        | 14.4 Mbps (DL)                                      |
| Shared Channel   | HS-DSCH — time-multiplexed among users              |
| Scheduling       | Moved to NodeB (2ms TTI — fast, channel-aware)      |
| AMC              | Adaptive Modulation & Coding (QPSK → 16QAM → 64QAM) |
| HARQ             | Hybrid ARQ with soft combining at NodeB             |
| No Power Control | Transmits at full power, adapts rate instead        |

## Release 6 – HSUPA (High Speed Uplink Packet Access)

| Feature    | Improvement                                   |
|------------|-----------------------------------------------|
| Peak Rate  | 5.76 Mbps (UL)                                |
| Channel    | E-DCH (Enhanced Dedicated Channel)            |
| Scheduling | NodeB-controlled with 2ms/10ms TTI            |
| HARQ       | Uplink soft combining                         |
| Benefit    | Better latency, higher throughput for uploads |

## Release 7/8 – HSPA+

- 64QAM downlink, 16QAM uplink
- MIMO (2×2) for downlink
- Dual-carrier operation (2 × 5 MHz)
- Peak rates: **42 Mbps DL, 11.5 Mbps UL**
- **Flat architecture option:** Direct NodeB-to-GGSN (bypassing RNC)

## The Shift in Philosophy

R99: RNC controls everything → Centralized, slow adaptation  
R5+: NodeB makes fast decisions → Distributed, rapid adaptation (2ms vs 100ms)

This shift foreshadowed LTE's design where the eNodeB handles almost all radio decisions.

## 3.7 IMS (IP Multimedia Subsystem) – Introduction

### What is IMS?

IMS is a **framework for delivering IP-based multimedia services** over mobile networks. Introduced in 3GPP Release 5, it represents the vision of converging ALL services (voice, video, messaging, presence) onto IP.

### Why IMS?

| Problem                                       | IMS Solution                                     |
|-----------------------------------------------|--------------------------------------------------|
| CS domain can't evolve (legacy SS7)           | All-IP signaling using SIP                       |
| Adding new services requires network upgrades | Application servers plug in modularly            |
| Voice and data are separate worlds            | Unified session control for all media            |
| No service interaction (voice OR data)        | Combinational services (voice + video + sharing) |

### Key IMS Components

- **P-CSCF (Proxy-CSCF)**: First contact point, SIP proxy, policy enforcement
- **I-CSCF (Interrogating-CSCF)**: Entry point from other networks, routes to correct S-CSCF
- **S-CSCF (Serving-CSCF)**: Core session control, service trigger logic, SIP registrar
- **HSS (Home Subscriber Server)**: Evolution of HLR, stores user profiles and service triggers
- **Application Servers**: Provide actual services (VoIP, conferencing, messaging)

### IMS Signaling Protocol: SIP

IMS uses **SIP (Session Initiation Protocol)** instead of SS7:

- Text-based, extensible
- Works natively over IP
- Supports any media type (voice, video, gaming, IoT)
- Enables service composition and mashups

### IMS's Role in the 2G→4G Journey

|                           |                     |                           |
|---------------------------|---------------------|---------------------------|
| 2G: Voice = CS (SS7)      | Data = None/CSD     |                           |
| 3G: Voice = CS (SS7)      | Data = PS (GTP)     | IMS = Optional overlay    |
| 4G: Voice = VoLTE via IMS | Data = PS (GTP)     | IMS = Mandatory for voice |
| 5G: Voice = VoNR via IMS  | Data = PS (GTP/SBA) | IMS = Still essential     |

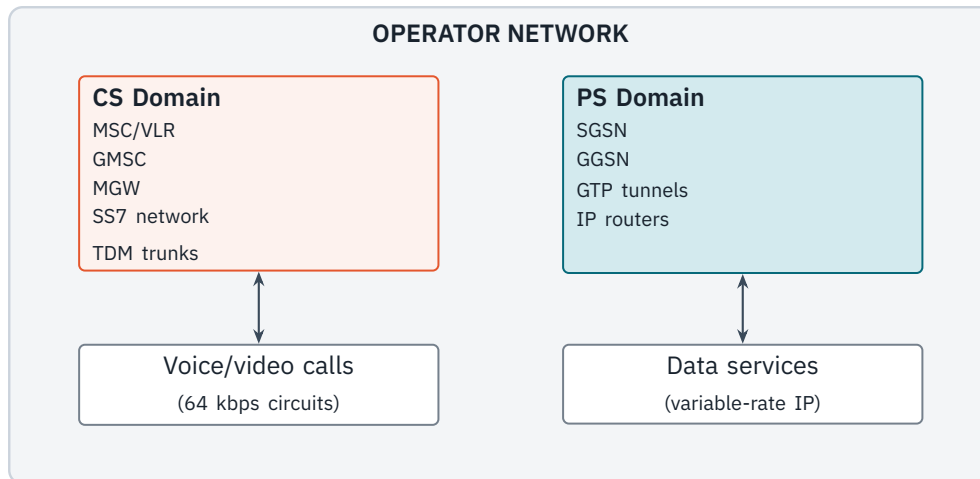
### 3.8 Major Comparison Table: Mobile Network Generations

| Feature                    | 2G (GSM)                                  | 2.5G (GPRS)                                               | 3G (UMTS R99)                                         | 3.5G (HSPA)                                          | 4G (LTE)                                                     |
|----------------------------|-------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------|
| <b>Radio Access</b>        | TDMA/FDMA (200 kHz channels, 8 timeslots) | Same as GSM (shared timeslots)                            | WCDMA (5 MHz carrier, spreading codes)                | WCDMA + shared channels (HS-DSCH/E-DCH)              | OFDMA DL / SC-FDMA UL (1.4–20 MHz)                           |
| <b>Core Network</b>        | MSC/VLR/HLR (SS7-based)                   | MSC + SGSN/ GGSN added                                    | MSC/VLR (CS) + SGSN/GGSN (PS)                         | Same as 3G (optional flat arch in HSPA+)             | EPC: MME/ S-GW/P-GW (all-IP, flat)                           |
| <b>Switching Type</b>      | Circuit-switched only                     | CS (voice) + PS (data)                                    | CS (voice) + PS (data)                                | CS (voice) + PS (data)                               | Packet-switched ONLY                                         |
| <b>Peak Data Rate (DL)</b> | 9.6 kbps (CSD) / 14.4 kbps (HSCSD)        | 171.2 kbps (theoretical) / ~40 kbps (practical)           | 384 kbps (pedestrian) / 2 Mbps (indoor)               | 14.4 Mbps (HSDPA) / 42 Mbps (HSPA+)                  | 300 Mbps (Cat 5) / 3 Gbps (Cat 20)                           |
| <b>Peak Data Rate (UL)</b> | 9.6 kbps                                  | 171.2 kbps (theoretical) / ~20 kbps (practical)           | 384 kbps                                              | 5.76 Mbps (HSUPA) / 11.5 Mbps (HSPA+)                | 75 Mbps (Cat 5) / 1.5 Gbps (Cat 20)                          |
| <b>Latency (RTT)</b>       | ~500 ms (data)                            | ~300–600 ms                                               | ~150–200 ms                                           | ~50–100 ms (HSPA) / ~30 ms (HSPA+)                   | ~10–30 ms (user plane) / ~50 ms (control)                    |
| <b>Mobility Support</b>    | Handover (hard) up to 250 km/h            | Same as GSM                                               | Soft handover + hard handover, up to 500 km/h         | Same as UMTS                                         | Hard handover only, up to 500 km/h                           |
| <b>Signaling Protocol</b>  | SS7 (MAP, ISUP, BSSAP)                    | SS7 + GTP-C                                               | SS7 (RANAP, MAP) + GTP-C                              | Same as 3G                                           | Diameter + GTP-C v2 (all IP-based)                           |
| <b>Main Innovation</b>     | Digital voice, SMS, roaming, SIM          | Always-on data, IP connectivity, shared radio             | Wideband data, video calls, multimedia, soft handover | Fast scheduling at NodeB, AMC, HARQ, shared channels | All-IP flat architecture, OFDMA, MIMO, VoLTE                 |
| <b>Main Limitation</b>     | No real data capability                   | Low throughput, high latency, shared with voice timeslots | Dedicated channels waste resources for bursty data    | Still CS voice, complex dual-stack                   | No native voice (needs VoLTE/IMS), complex handover to 2G/3G |

## 3.9 The CS/PS Split – And Why LTE Killed It

### ■ The Problem with Dual Domains

Running CS and PS domains in parallel meant:



**Figure 3.5** Two parallel core domains in the operator network

### ■ Costs of the Split Architecture

| Problem                  | Impact                                                                     |
|--------------------------|----------------------------------------------------------------------------|
| Duplicate infrastructure | Two complete core networks to maintain, upgrade, and staff                 |
| Inefficient spectrum use | CS reserves capacity even during silence (60% of a call is silence)        |
| No service convergence   | Voice and data can't easily interact (e.g., sharing a photo during a call) |
| SS7 is legacy            | Hard to extend, proprietary equipment, no IP ecosystem benefits            |
| Separate billing         | Voice minutes + data MB = complex charging systems                         |
| Scaling limitations      | CS scales with ports/trunks (expensive), PS scales with bandwidth (cheap)  |

### ■ Why LTE Unified to Packet-Only (EPC)

LTE's Evolved Packet Core (EPC) made a radical decision: **NO circuit-switching at all.**

#### ■ The Technical Enablers (What Changed by 2008):

1. **IP Networks Matured:** QoS mechanisms (DiffServ, MPLS, dedicated bearers) could now guarantee voice quality

2. **Codec Efficiency:** AMR-WB and EVS codecs work well over IP with minimal bandwidth (~12 kbps)
3. **Low Latency Achieved:** LTE's ~10ms RTT is actually BETTER than CS voice (~50ms codec delay anyway)
4. **VoIP Proven:** Skype, Vonage proved IP voice was commercially viable
5. **IMS Was Ready:** After years of development, IMS could replace SS7 call control with SIP
6. **Economics:** IP equipment costs 1/10th of equivalent TDM/SS7 equipment

### ■ The EPC Architecture (All Packet):

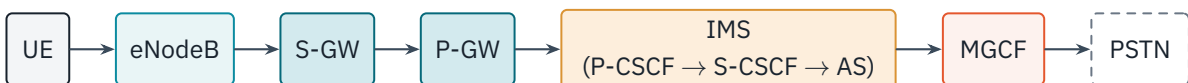
| Element | Replaces            | Function                                                     |
|---------|---------------------|--------------------------------------------------------------|
| MME     | MSC (control plane) | Mobility, authentication, bearer management — signaling only |
| S-GW    | SGSN (user plane)   | Local mobility anchor, data forwarding between eNodeBs       |
| P-GW    | GGSN                | External network gateway, IP allocation, policy/charging     |
| HSS     | HLR + AuC           | Subscriber data, authentication vectors                      |
| PCRF    | — (new)             | Policy and Charging Rules — enables QoS per-flow             |

### ■ How Voice Works Without CS (VoLTE):

Traditional 3G voice — dedicated 64 kbps circuit



LTE voice (VoLTE) — IP packets on a QCI 1 dedicated bearer, about 40 kbps AMR-WB



**Figure 3.6** Voice path in 3G (circuit) versus LTE (VoLTE over IMS)

### ■ Benefits of Unified Packet Architecture:

1. **Single infrastructure** to maintain → 50%+ OPEX reduction
2. **Statistical multiplexing** → no wasted capacity during silence
3. **Service convergence** → voice + video + data on same bearer simultaneously
4. **Rapid service innovation** → new services via Application Servers, not network upgrades
5. **HD Voice** → AMR-WB (wideband) sounds dramatically better than GSM's narrowband codec
6. **Network simplification** → fewer interfaces, fewer protocol stacks, fewer points of failure

## ■ The Trade-off

| Aspect            | CS (2G/3G)                        | PS-only (LTE)                                  |
|-------------------|-----------------------------------|------------------------------------------------|
| Voice reliability | Extremely high (30+ years proven) | Depends on coverage and VoLTE implementation   |
| Call setup time   | ~3–5 seconds                      | ~1–2 seconds (SIP is faster)                   |
| Emergency calls   | Always works (CS fallback)        | Requires VoLTE or CSFB to 2G/3G                |
| Complexity        | Simple for voice, complex overall | Simple overall, complex for voice QoS          |
| Coverage gaps     | Voice still works at cell edge    | Voice drops if LTE coverage drops (needs CSFB) |

**CSFB (Circuit-Switched Fallback):** Early LTE networks fell back to 3G/2G for voice calls. This was a transitional solution until VoLTE coverage was ubiquitous.

### 3.10 Summary: Key Takeaways

1. **3G (UMTS)** introduced WCDMA radio with dedicated channels and maintained the 2G CS/PS split core
2. **UTRAN** separates radio functions: NodeB (physical layer) and RNC (radio resource management)
3. **The Iu interface** cleanly separates radio access from core — enabling independent evolution
4. **HSPA (R5/R6)** moved intelligence to NodeB (fast scheduling, HARQ, AMC) — a preview of LTE’s philosophy
5. **IMS** provides the all-IP service layer that eventually replaces SS7 for voice
6. **LTE unified everything to packets** because IP matured enough to guarantee voice QoS, and the economics of maintaining dual infrastructure became unjustifiable
7. **The evolution was gradual:** each generation solved real limitations of the previous one

#### Review Questions

1. Explain why UMTS maintained separate CS and PS domains instead of going all-packet from the start.
2. What is the difference between a Serving RNC and a Drift RNC?
3. How does HSDPA’s NodeB-based scheduling improve over R99’s RNC-based scheduling?
4. Draw the protocol stack for the Iu-PS interface.
5. Why was IMS necessary for LTE to eliminate the CS domain?
6. Compare the role of SGSN in 3G with S-GW in LTE — what changed and why?

*Module 3 — Advanced Communication Networks — 2G/3G Core Networks*

*Next: Module 4 — LTE/EPC Architecture*